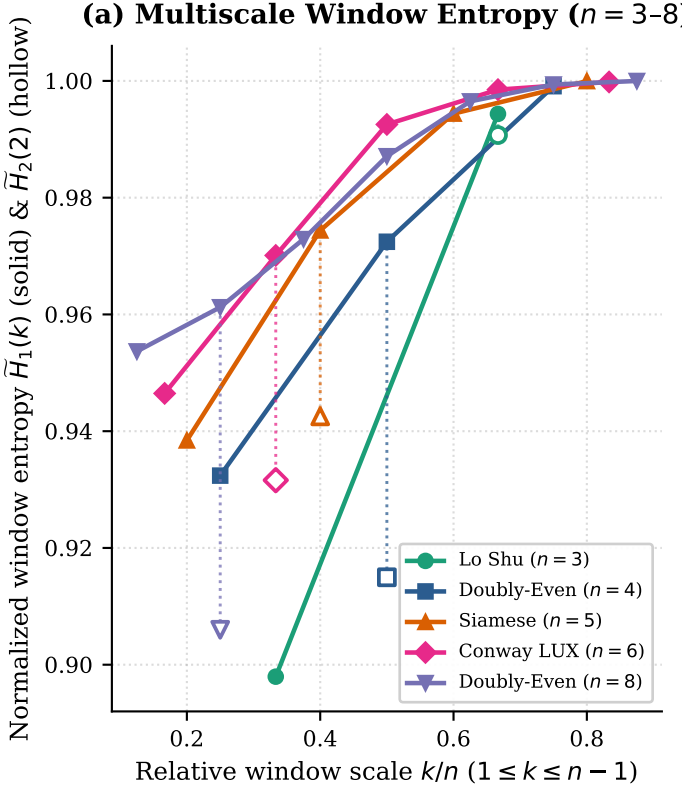
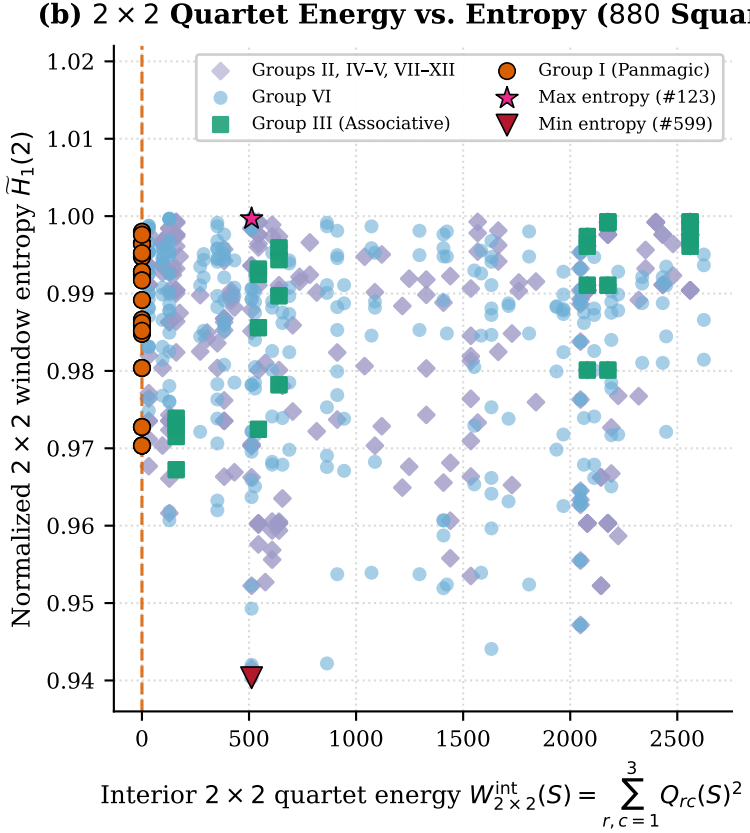
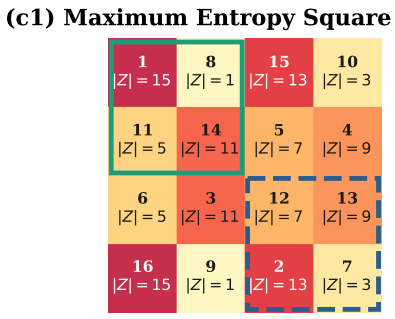
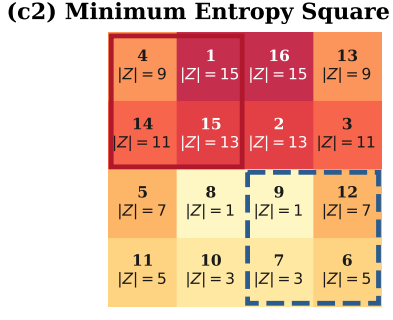

(a) Multiscale Window Entropy ($n = 3$–8)
Normalized window entropy $\widetilde{H}_1(k)$ (solid) & $\widetilde{H}_2(2)$ (hollow)
Relative window scale k/n $(1 \le k \le n-1)$
Lo Shu ($n = 3$)
Doubly-Even ($n = 4$)
Siamese ($n = 5$)
Conway LUX ($n = 6$)
Doubly-Even ($n = 8$)


(b) 2 × 2 Quartet Energy vs. Entropy (880 Squares)
Normalized 2 × 2 window entropy $\widetilde{H}_1(2)$
Interior 2 × 2 quartet energy $W^{\text{int}}_{2\times 2}(S) = \sum_{r,c=1}^{3} Q_{rc}(S)^2$
Groups II, IV–V, VII–XII
Group VI
Group III (Associative)
Group I (Panmagic)
Max entropy (#123)
Min entropy (#599)


(c1) Maximum Entropy Square (#123)
1 |Z| = 15
8 |Z| = 1
15 |Z| = 13
10 |Z| = 3
11 |Z| = 5
14 |Z| = 11
5 |Z| = 7
4 |Z| = 9
6 |Z| = 5
3 |Z| = 11
12 |Z| = 7
13 |Z| = 9
16 |Z| = 15
9 |Z| = 1
2 |Z| = 13
7 |Z| = 3


(c2) Minimum Entropy Square (#599)
4 |Z| = 9
1 |Z| = 15
16 |Z| = 15
13 |Z| = 9
14 |Z| = 11
15 |Z| = 13
2 |Z| = 13
3 |Z| = 11
5 |Z| = 7
8 |Z| = 1
9 |Z| = 1
12 |Z| = 7
11 |Z| = 5
10 |Z| = 3
7 |Z| = 3
6 |Z| = 5